



## Daehee Park

Assistant Professor

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

Department of Electrical Engineering and Computer Science

+82-10-3290-2155

dhpark@dgist.ac.kr

Google Scholar

Personal Website

ISL Lab Website

### RESEARCH INTERESTS

- **Multimodal Models for Physical AI:** Learning from vision, language, and motion for embodied understanding and action.
- **World Models and Motion Forecasting:** Modeling 3D environments, human motion, and multi-agent interactions to predict future states.
- **Prediction and Planning for Autonomous Systems:** Integrating perception, prediction, and planning for autonomous driving and robotics.
- **Efficient and Adaptive AI:** Model compression, model merging, and test-time adaptation for reliable deployment in dynamic environments.

### WORK EXPERIENCE

- **DGIST (Daegu Gyeongbuk Institute of Science and Technology)**  
Assistant Professor, Dept. of Electrical Engineering and Computer Science 2025.03 – Present  
Assistant Professor, Dept. of Interdisciplinary AI 2025.03 – Present
- **Qualcomm**  
Interim Engineering Intern, QCT Auto AD 2024.04 – 2024.09
- **Naver Labs**  
Research Intern, Autonomous Driving Team 2021.06 – 2021.08

### EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)**  
Ph.D. in Mechanical Engineering 2020.03 – 2025.02  
Advisor: Prof. Kuk-Jin Yoon
- **KAIST (Korea Advanced Institute of Science and Technology)**  
M.S. in Mechanical Engineering 2018.03 – 2020.02
- **KAIST (Korea Advanced Institute of Science and Technology)**  
B.S. in Mechanical Engineering 2013.03 – 2018.02  
Double Major in Business and Technology Management

### PUBLICATIONS

\* equal contribution, † corresponding author

- **[c17]** Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction  
Donghyeon Soon\*, Jiil Yoon\*, Hyeondong Kim, **Daehee Park**<sup>†</sup>  
*Neural Information Processing Systems (NeurIPS)*, 2026.
- **[c16]** SRA: Spatial Reasoning Adapter via Evolving Social Interaction Graphs for Trajectory Prediction  
Jaewoo Jeong\*, Seonkyu Song\*, Hyeonwoo Park, Jegyeong Cho, Yujin Bae, Giwon Lee, **Daehee Park**,  
Kuk-Jin Yoon<sup>†</sup>  
*Neural Information Processing Systems (NeurIPS)*, 2026.
- **[c15]** Unified Prediction and Planning via Conflict-Aware Disjoint Parameter Training  
Taewon Seo\*, Seonae Jeon\*, Giwon Lee\*, Kuk-Jin Yoon<sup>†</sup>, and **Daehee Park**<sup>†</sup>  
*European Conference on Computer Vision (ECCV)*, 2026.
- **[j2]** CascadeOcc: Rethinking 3D Occupancy World Models with Cascaded VQ Representations  
Kyumin Hwang\*, Wonhyeok Choi\*, Jaeyul Kim, Jihun Park, **Daehee Park**<sup>†</sup>, Sunghoon Im<sup>†</sup>  
*IEEE Signal Processing Letters*, 2026.

- **[c14]** EggHand: A Multimodal Foundation Model for Egocentric Hand Pose Forecasting  
Jaeyoung Choi\*, Hyeondong Kim\*, Yujin Kim, **Daehee Park**<sup>†</sup>  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, 2026.
- **[c13]** BitTP: The Lightweight Trajectory Prediction Model with BitLLM for Edge-Devices  
Mincheol Kang\*, HyunJin Lim\*, Bomin Kang\*, **Daehee Park**<sup>†</sup>  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, 2026.
- **[c12]** Generative Active Learning for Long-tail Trajectory Prediction via Controllable Diffusion Model  
**Daehee Park**, Monu Surana, Pranav Desai, Ashish Mehta, Reuben M. V. John, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- **[c11]** Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning  
Giwon Lee\*, Wooseong Jeong\*, **Daehee Park**, Jaewoo Jeong, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025 [[Highlight](#)].
- **[c10]** Non-differentiable Reward Optimization for Diffusion-based Autonomous Motion Planning  
Giwon Lee\*, **Daehee Park**\*, Jaewoo Jeong\*, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- **[p1]** Self-Supervised 3D Occupancy Prediction with Temporal Consistency  
Jae-Seok Jeong, Sung-Hoon Yoon, **Daehee Park**, Kuk-Jin Yoon<sup>†</sup>.  
*arXiv preprint*, 2025.
- **[c9]** Multi-modal Knowledge Distillation-based Human Trajectory Forecasting  
Jaewoo Jeong, SeoHee Lee, **Daehee Park**, Giwon Lee, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- **[c8]** Diffusion-Guided Weakly Supervised Semantic Segmentation  
Sung-Hoon Yoon, Hoyong Kwon\*, Jaeseok Jeong\*, **Daehee Park**, Kuk-Jin Yoon<sup>†</sup>  
*European Conference on Computer Vision (ECCV)*, 2024.
- **[c7]** T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory  
**Daehee Park**, Jae-Seok Jeong, Sung-Hoon Yoon, Jaewoo Jeong, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- **[c6]** Multi-agent Long-term 3D Human Pose Forecasting via Interaction-aware Trajectory Conditioning  
Jaewoo Jeong\*, **Daehee Park**\*, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [[Highlight](#)].  
[Winner, Best Paper Award at Qualcomm Innovation Fellowship Korea](#)
- **[c5]** Improving Transferability for Cross-domain Trajectory Prediction via Neural Stochastic Differential Equation  
**Daehee Park**, Jaewoo Jeong, Kuk-Jin Yoon<sup>†</sup>  
*AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
- **[c4]** Leveraging Future Relationship Reasoning for Vehicle Trajectory Prediction  
**Daehee Park**, Hobin Ryu, Yunseo Yang, Jegyeong Cho, Jiwon Kim, Kuk-Jin Yoon<sup>†</sup>  
*International Conference on Learning Representations (ICLR)*, 2023.  
[1st Place, nuScenes Prediction Challenge](#)
- **[c3]** BIPS: Bi-modal Indoor Panorama Synthesis via Residual Depth-Aided Adversarial Learning  
Changgyoon Oh\*, Wonjune Cho\*, Yujeong Chae\*, **Daehee Park**\*, Lin Wang, Kuk-Jin Yoon<sup>†</sup>  
*European Conference on Computer Vision (ECCV)*, 2022.  
[Grand Prize, Best Paper Award at Workshop on Image Processing and Image Understanding 2022](#)
- **[c2]** Unlocking the Potential of Ordinary Classifier: Class-specific Adversarial Erasing Framework for Weakly Supervised Semantic Segmentation  
Hyeokjun Kweon\*, Sung-Hoon Yoon\*, Hyeonseong Kim, **Daehee Park**, Kuk-Jin Yoon<sup>†</sup>  
*IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

- [j1, c1] Identifying Reflected Images from Object Detector in Indoor Environment Utilizing Depth Information  
**Daehee Park**, Yong-Hwa Park<sup>†</sup>  
*IEEE Robotics and Automation Letters (RA-L) and IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

## PATENTS

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- 태스크 간 충돌을 완화하는 파라미터 분리 학습 기반 모델 병합 방법 및 장치  
**Korean Patent**, Provisional application, 10-2026-0133460, 2026.07.21.
- 궤적 예측에서 거대언어모델 경량화를 위한 비트 양자화 기법  
**Korean Patent**, 10-2026-0093495.
- 1인칭 시점 손 동작의 인지, 예측 및 제어를 위한 멀티모달 파운데이션 모델 프레임워크 및 최적화 방법  
**Korean Patent**, 10-2025-0209315.
- Electronic device and method of identifying false image of object attributable to reflection in indoor environment thereof  
**United States Patent**, Registered, US11282178B2.
- Dispositif électronique et procédé d'identification de fausse image ou de faux objet imputable à la réflexion dans un environnement intérieur correspondant  
**European Patent**, Registered, EP3772701.
- 전자 장치 및 그의 실내 환경에서 반사에 의한 객체 허상을 식별하기 위한 방법  
**Korean Patent**, Registered, KR10-2287478-0000.

## POSITIONS OF RESPONSIBILITY

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- Planning Committee Member, 2027 Physical AI R&D Projects, IITP, 2026.09
- Program Committee Member, IPIU 2027
- Planning Committee Member, 2026 Culture, Sports and Tourism R&D Projects, KOCCA, 2025.09

## AWARDS AND HONORS

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- Winner, NVIDIA Academic Grant Program, NVIDIA 2025.12
- Outstanding Doctoral Graduate Award, KAIST ME 2025.12
- Winner, Qualcomm Innovation Fellowship Korea (QIFK) 2024.12
- Research Scholarship, Samsung Advanced Institute of Technology 2022.09 – 2025.02
- 1st Place, Best Paper Awards, IPIU 2022
- Outstanding Achievement Award, KAIST ME 2016.09

## ACADEMIC REVIEWING & SERVICE

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- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE/CVF International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- AAAI Conference on Artificial Intelligence (AAAI)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

- IEEE Intelligent Vehicles Symposium (IV)
- Winter Conference on Applications of Computer Vision (WACV)
- ACM International Conference on Multimedia (ACM MM)
- British Machine Vision Conference (BMVC)
- Pattern Recognition (PR)
- IEEE Transactions on Intelligent Vehicles (T-IV)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Engineering Applications of Artificial Intelligence (EAAI)

## RESEARCH GRANTS

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- 숙련자 시연데이터 기반 조선소 작업 특화 양팔 모바일 매니플레이터 및 Physical AI 작업스킬 실증 플랫폼 개발  
(PI) 산업통상자원부 / 한국산업기술기획평가원 2026.08 - 2030.12
- 도시형 생성 지능을 위한 사회적 군중 시뮬레이션 기술 개발  
과학기술정보통신부 / 인공지능산업융합사업단 (H100 x 8ea) 2026.07 - 2027.01
- 희소 궤적 월드 액션 모델을 활용한 양손 조작 행동 생성 기술 개발  
과학기술정보통신부 / 한국전자기술연구원 (H100 x 8ea) 2026.07 - 2026.11
- 경량형 온디바이스 인식 모델 최적화 연구 개발  
(PI) 경상북도 / 한국전자통신연구원 2026.05 - 2027.12
- 한국형 Physical LLM 의 실시간성과 안정성을 위한 가변형 추론 프레임워크 개발  
한국정보통신진흥협회 / 정보통신산업진흥원 (B200 x 16ea) 2026.05 - 2026.06
- 다중 휴머노이드의 자율 협업을 위한 비전-언어 모델 기반 적응형 장기 · 복합작업 계획 기술  
과학기술정보통신부 / 한국연구재단 2026.04 - 2030.12
- 아카이브 영상 콘텐츠 활용을 위한 AI Outpainting 및 콘텐츠 변환 기술 개발  
(PI) 한국콘텐츠진흥원 2026.04 - 2028.12
- 자기 진화형 Physical LLM: 불확실한 환경에서 행동 파운데이션 모델의 강건하고 효율적인 적응 기술 연구  
(PI) 과학기술정보통신부 / 한국연구재단 2026.03 - 2030.02
- NVIDIA Academic Grant  
(PI) NVIDIA (Pro6000 x 4ea) 2026.02 - 2026.08
- 행동 파운데이션 모델의 동적 환경으로 확장을 통한 안전하고 범용적인 인간-로봇 협동 연구  
(PI) 과학기술정보통신부 / 한국연구재단 2025.09 - 2026.08
- 로봇및기계전자공학연구소  
교육부 / 한국연구재단 2025.09 - 2034.08
- Physical AI 분야의 거대 생성모델 기술 선도를 위한 LG AI STAR 인재양성 사업  
(PI) 과학기술정보통신부 / 정보통신기획평가원 2025.07 - 2028.12
- (InnoCORE) 초거대언어모델 혁신 연구단  
과학기술정보통신부 / 한국과학기술원 2025.07 - 2029.12
- (InnoCORE) 바이오 체화형 피지컬 AI 연구단  
과학기술정보통신부 / DGIST 2025.07 - 2029.12
- 초거대 클라우드팜 AI GPU 가상자원  
경북IT융합 산업기술원 (H100 x 2ea) 2025.07 - 2025.12
- 생성형 AI 기반 16bit HDR 솔루션 개발  
삼성전자-KAIST 산학과제 2025.05 - 2027.04
- AI스타펠로우십지원(대구경북과학기술원)  
과학기술정보통신부 / 정보통신기획평가원(IITP) 2025.04 - 2028.12

## INVITED TALKS

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- SMIT Pathfinding Seminar, Samsung Electronics Semiconductor R&D Center *2026.07.31*  
Advances in Deep Learning Model Adaptation
- IEIE Image Understanding Summer School, Sookmyung Women's University *2026.07.15*  
Research and Paper Writing in Computer Vision and Machine Learning from a Non-CS Background
- Young Leaders Session (YLS), IEIE Summer Conference *2026.06.24*  
Self-Evolving Physical AI for Robust Decision Making
- KETI, Pangyo *2026.03.20*  
Motion Prediction and Its Application for Physical AI
- ICT 4.0 Special Lecture, Hanbat National University *2025.11.24*  
Prediction and Planning for Physical AI in Robotics & Autonomous Driving
- New Faculty Lunchtime Seminar, DGIST *2025.10.28*  
Decision Making in Physical AI
- Young Researcher Talk, AISP 2025, Kyungpook National University *2025.09.26*  
Prediction and Planning for Physical AI in Robotics and Autonomous Driving
- ETRI, Daegu-Gyeongbuk Research Center *2025.09.17*  
Decision-making AI under Uncertainty: A Prediction and Planning Perspective
- EECS & AI Colloquium, DGIST *2025.09.09*  
Prediction and Planning for Physical AI in Robotics & Autonomous Driving
- IEIE Joint Summer School on Image Understanding and Image Processing *2025.07.18*  
Toward Decision-making AI: A Prediction and Planning Perspective
- 8th Daegu AI Researchers' Forum Special Seminar, Keimyung University *2025.05.30*  
Data-driven Decision Making in Autonomous Systems: A Prediction and Planning Perspective
- KSAE Spring Conference, Convergence SW & AI Session *2025.05.22*  
T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory

## TEACHING

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Introduction to Deep Learning, Fall 2026 (in progress).  
Advanced Deep Learning, Spring 2026.  
Artificial Intelligence and Machine Learning Engineering (AT620), Spring 2026.  
Co-instructor with Prof. Sunghoon Im (50% teaching share).  
Artificial Intelligence Basics, Fall 2025.  
Computer Architecture, Spring 2025.  
Course development: Deep Learning (AI Microdegree), Fall 2026 (in development).